

Owen Garnier

Post-doctoral researcher in Mathematics

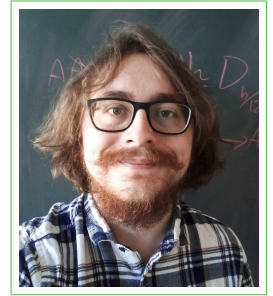
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Studies

2024 **PhD in mathematics**, LAMFA, Amiens.

Titre : Garside groupoids and complex braid groups.

Director : Ivan Marin

Other members of the Jury : Yves Guiraud, Jon McCammond, Adrien Brochier, Cédric Bonnafé, David Chataur, François Digne, Eddy Godelle.

2018–2021 **Masters degree in mathematics (Algebra, number theory and applications)**, Université de Picardie Jules Verne, Amiens.

With honors (Mention très bien).

Thesis: Théorie de Garside, applications aux groupes de tresses complexes. Cas de G_{31} . (supervisor: Ivan Marin).

2020 **Agrégation externe de mathématiques**, Rank: 10/387.

2018–2019 **Fourth year mathematics**, Université de Picardie Jules Verne, Amiens.

With honors (Mention très bien).

Thesis: Compléments sur l'homologie générale. Théorie des catégories (ex)triangulées. (supervisor: Yann Palu).

2015–2018 **Licence in mathematics**, Université de Picardie Jules Verne, Amiens.

With honors (Mention très bien).

Professional experiences

2025–2026 **Post-doctoral researcher**, University of Sevilla, Sevilla (Spain).

Supervised by María Cumplido Cabello.

2024–2025 **Temporary research and teaching assistant (ATER)**, Université de Picardie Jules Verne, Amiens (France).

2021–2024 **PhD student with teaching duties**, Université de Picardie Jules Verne, Amiens (France).

Supervised by Ivan Marin.

Distinctions and prizes

2025 **PhD thesis award of the Université de Picardie Jules Verne**, Award given to the two best PhD thesis in science, technology and health subjects defended in 2024

Research papers

- 2026 **Springer categories for regular centralizers in well-generated complex braid groups.**
Advances in Mathematics 495, Paper No. 110971.
- 2024 **Uniqueness of roots up to conjugacy in circular and hosohedral-type Garside groups.**
Journal of Group Theory 27, 5 (2024), pp. 1091–1128.
- 2024 **Generalization of the Dehornoy-Lafont order complex to categories. Application to exceptional braid groups.**
Applied Categorical Structures 32, 1 (2024) Paper No. 1, 28 pp.
- 2023 **Regular theory in complex braid groups.**
Journal of Algebra 620 (2023), pp. 534–557.

Preprints

- 2025 **Generalized J -groups, J -braid groups and Seifert link groups, With I. Haladjian**
- 2025 **Conjugacy invariants and rigidity in Garside groups: a uniformity phenomenon, With M. Calvez, J. González-Meneses and B. Wiest**
Submitted to the *Journal of Algebra*
- 2025 **Normalisers of parabolic subgroups of Artin-Tits groups and Tits cone intersections, With E. Heng, A. Licata and O. Yacobi**
Submitted to *Geometry and Topology*
- 2025 **Proof of Shvartsman's conjecture on braid groups of projective complex reflection groups.**
Accepted in the *Proceedings of the American Mathematical Society*
- 2024 **Parabolic subgroups of complex braid groups: the remaining case.**

Communications and Talks

Talks in conferences

- 2026 **Seifert link groups and their torsion quotients,**
Workshop on geometric topology, Sevilla (Spain).
- 2025 **Sous-groupes paraboliques des trickle groups,**
Workshop on trickle groups and their applications, Tours (France).
- Système de réécritures et solutions au problème du mot,**
Workshop on trickle groups and their applications, Tours (France).
- Sous-groupes paraboliques des groupes de tresses complexes,**
Journées jeunes chercheurs et chercheuses en topologie, Amiens (France).
- Groupeïdes de Garside et groupes de tresses complexes,**
Colloque tournant du RT algèbre, Clermont-Ferrand (France).
- 2023 **Homology computations for complex braid groups with the Dehornoy-Lafont order complex,**
Conférence 2023 du GDR Théorie de l'Homotopie et Applications, Lille (France).
- Bessis braid category for the complex braid group B_{31} ,**
Winterbraids XII, Tours (France).
- 2022 **Regular theory in complex braid groups,**
Journées tresses 2022, Amiens (France).

Talks in seminars

- 2025 **Normalisateurs des sous-groupes paraboliques dans les groupes d'Artin et de Coxeter,**
Topology seminar, IMJ-PRG (Paris, France).
- 2025 **Normalizers of parabolic subgroups in Artin and Coxeter groups,**
Group theory seminar, ICMAT (Madrid, Spain).
- Projective complex braid groups,**
Algebra and Topology seminar, ANU (Canberra, Australia).

- Sous-groupes paraboliques, des tresses aux groupoïdes de Garside,**
Geometry, Algebra, Dynamics and Topology seminar, IMB (Dijon, France).
- Groupoïdes de Garside paraboliques et applications aux groupes de tresses complexes,**
Algebra and Geometry seminar, LMNO (Caen, France).
- 2024 **Des partitions non-croisées à l'étude des sous-groupes paraboliques des groupes de tresses complexes,**
SPACE seminar, IDP (Tours, France).
- Sous-groupes paraboliques et éléments réguliers des groupes de tresses complexes,**
Darboux seminar, IMAG (Montpellier, France).
- Homology of categories and the Dehornoy-Lafont order complex,**
Seminar of the "Ouragan" team, IMJ-PRG (Paris, France).
- 2024 **Description de l'espace des orbites régulières d'un groupe de réflexions complexes,**
Topology seminar, LPP (Lille, France).
- 2023 **Étude Garside du groupe de tresses complexe $B(G_{31})$,**
GATo Seminar, LAMFA (Amiens, France).
- Garside study of the complex braid group $B(G_{31})$,**
Groupes, representations and geometry seminar, IMJ-PRG (Paris, France).
- Homology of a category and the Dehornoy-Lafont order complex,**
Paris algebra seminar, IHP (Paris, France).
- 2022 **Regular theory in complex braid groups,**
Algebra seminar, Dep. de Álgebra (Seville, Espagne).
- Théorie régulière dans les groupes de tresses complexes,**
GATo seminar, LAMFA (Amiens, France).

Conferences

- 2026 **Workshop on geometric topology,**
Seville (Spain), 19/02.
- 2025 **Workshop on trickle groups and their applications,**
Tours (France), from 25/06 to 27/06.
- Artin groups, Braids and mapping class groups: Celebrating the work of Luis Paris,**
Cáceres (Spain), from 16/06 to 20/06 .
- Journées jeunes chercheurs et chercheuses en topologie,**
Amiens (France), from 09/04 to 11/04.
- Colloque tournant du RT algèbre,**
Clermont-Ferrand (France), from 05/03 to 07/03.
- Winterbraids XIV,**
Bordeaux, from 03/02 to 06/02.
- 2023 **Conférence 2023 du GDR Théorie de l'Homotopie et Applications,**
Lille (France), from 24/10 to 27/10.
- École d'été: À la découverte des correspondances de Langlands locales,**
Amiens (France), du 28/08 au 01/09.
- Workshop: Various Guises of Reflection Arrangements,**
Edimbourg (UK), from 13/03 to 17/03.
- Spring School on Real, Complex, and Symplectic Reflection Groups,**
Bochum (Germany), from 06/03 to 10/03.
- Winterbraids XII,**
Tours, from 21/02 to 24/02.
- 2022 **Journées tresses 2022 : Groupes de tresses généralisés,**
Amiens (France), from 29/08 to 02/09.

- 2021 **Summer school: The dual approach to Coxeter and Artin groups; Garside theory and applications**,
Berlin (Germany), from 30/08 to 04/09.

Teaching

- 2024–2025
- Calculus (first year science) (30h).
 - Set theory (second year mathematics) (20h).
 - Groups, Rings, Fields (third year mathematics) (36h).
 - Linear algebra 1 (first year science) (32h).
 - parametrized curves (first year mathematics) (36h).
 - Complex analysis (third year mathematics) (20h).
- 2023–2024
- Advanced linear algebra (third year mathematics) (20h).
 - Complex analysis (third year mathematics) (20h).
 - Preparation to the “agrégation externe” (M.Sc. mathematics) (24h).
- 2022–2023
- Advanced linear algebra (third year mathematics) (20h).
 - Algebraic algorithmics (third year mathematics) (20h).
 - Preparation to the “agrégation externe” (M.Sc. mathematics) (24h).
- 2021–2022
- Calculus (first year science) (24h).
 - Advanced linear algebra (third year mathematics) (20h).
 - Complex numbers and geometry (third year mathematics) (20h).

Other activities

- 2025 **Co-organizer of the workshop on trickle groups and their applications**, Tours.
With Igor Haladjian.
- 2023–2024 **Co-organizer of the PhD student seminar of the LAMFA**, Amiens.
With Nicolas Beuvin et Michaël Schoonheere.
- 2022–2023 **Co-organizer of the PhD student seminar of the LAMFA**, Amiens.
With Felipe Arbulú et Ismaïl Razack.
- 2022 **Research stay in Sevilla**, *Universidad de Sevilla*, Sevilla (Spain).
Work with Juan Gonzalez-Meneses (3 weeks).
- 2022 **Co-organizer of the braid meeting 2022**, Amiens.
With Ivan Marin and Karine Sorlin.